

Master Thesis - MSc. Finance and Investments

The Effect of Combined Acquirer Information Signaling on Cross-Border M&A Performance

Date of Submission: June 7, 2024

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Preface

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Acknowledgements

I thank my coach and co-reader, Stefan van Kampen and Ahmet Karpuz, for their constructive feedback and helpful answers during my work on this thesis.

Abstract

In this research paper, I investigate the impact of combined acquirer information signaling on cross-border mergers and acquisitions (M&A) performance. Payment method and private equity (PE) funding of the acquiring firm are the information signals investigated. In this event study methodology, M&A performance is measured using cumulative abnormal returns (CAR) surrounding the announcement date. Based on a sample of 465 corporate deals and various ordinary least squares (OLS) regressions, I find evidence to suggest that the takeover's payment method weakly significantly impacts M&A performance, with an impact on CAR relative to stock payment of 3.064%. In addition, there is no evidence that indicates a statistically significant relationship between PE funding and M&A performance, contrary to findings from Humphery-Jenner et al. (2017). Finally, the combined impact of the two signals on CAR is not shown to differ significantly from the sum of their individual impacts. Consequently, the study reveals no evidence in support of a meaningful relationship between combined acquirer information signaling and M&A performance. The results of this research contribute to the literature by addressing a well-established need to investigate combinations of M&A performance drivers.

Keywords: M&A performance, information asymmetries, information signaling, payment, PE funding.

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1. Introduction

Mergers and acquisitions (M&A) present an extremely valuable tool for companies seeking to access growth opportunities, be it through gaining access to new resources, or expanding market reach (Calipha et al., 2010). The potential of corporate takeovers to create long-lasting value is underlined by the substantiality of M&A markets; in 2023 alone, over 50,000 deals were struck worldwide, totaling more than \$2,000 billion in value (Levy, 2024).

It is important, however, for firms to execute takeovers with care, as to fully capitalize on potential synergies (Agrawal et al., 2017). Despite the magnitude of this market, global M&A activity witnessed significant downturns in recent years, leading to a record-low deal value this decade (Fontanella-Khan, 2024). Nevertheless, recent forecasts by industry experts suggest a rebound in deal volume and value from 2024 onwards (Fontanella-Khan, 2024; Levy, 2024). Thus, M&A remains highly relevant for academic inquiry and professional management.

The literature on M&A typically organizes into two main areas: motivators of M&A activity; and moderators of M&A performance (Haleblian et al., 2009). Motivators of M&A activity describe the reasons for firms to engage in takeovers (Haleblian et al., 2009). Moderators of M&A performance explore the relationship between acquisition activity and success (Haleblian et al., 2009). Broadly speaking, scholars commonly classify between acquisitions driven by strategic or financial intent (Osuma et al., 2021). More specifically, frequently cited reasons to acquire include gaining market power, enhancing efficiency, and managerial incentives such as compensation (Haleblian et al., 2009). Regarding the moderators of M&A performance, a broad distinction is made between firm-level and deal-level characteristics (Haleblian et al., 2009). Examples of frequently studied performance drivers are deal types, referring to the nature of the acquirer's offer, ownership structures, executive compensation mechanisms, and the composition of the board of directors (Defrancq et al., 2021; Haleblian et al., 2009).

Yet, despite the widespread popularity of acquisitions, many takeovers fail to create positive value. In fact, the literature on M&A performance presents a wide landscape of contradictions, with many empirical studies suggesting that acquisitions fail to generate significant positive value for acquiring firms, despite often yielding positive returns for targets (e.g. Haleblian et al., 2009; King

et al., 2004; King et al., 2021). This notion of value destruction in M&A is consistently highlighted by insights from both academic and professional literature (e.g. Calipha et al., 2010; Christensen et al., 2011; Reuer, 2005; Selden & Colvin, 2003). As a result, there is ample need for continued research into the performance drivers of M&A, seeing as current research fails to properly explain the variance in performance across acquisitions (Cartwright & Schoenberg, 2006; Haleblan et al., 2009; King et al., 2004; King et al., 2021).

Moreover, existing research remains reductionist in its approach, primarily examining performance drivers in isolation (Haleblan et al., 2009). Various scholars have emphasized the need to integrate knowledge by exploring interactions among previously identified antecedents to determine their relative significance (Haleblan et al., 2009; King et al., 2004; King et al., 2012). Consequently, an interesting research opportunity is to explore combinations of M&A performance drivers.

Information asymmetries are commonly cited as a primary deterrent of M&A performance, particularly in the context of cross-border M&A due to country-level differences (Erel et al., 2012; Erel et al., 2022; Gomes et al., 2013; Tanna et al., 2021). Differences in information typically arise between the target and acquirer, or between the acquirer's management and its shareholders (Bergh et al., 2019; Boeh, 2011; Moeller & Schlingemann, 2005; Song et al., 2021). As a result, a large portion of cross-border M&A research aims to discover how to reduce these issues (Boeh, 2011; Erel et al., 2022).

One regularly referenced approach to reducing information asymmetries is information signaling (Gale, 2001; Michaely & Shaw, 1994; Stiglitz, 2002). Signaling refers to firm-level or deal-level characteristics of the target or acquiring firm that may signal the quality of the investment to shareholders of acquiring firms. Effective signals convey information about the quality of the deal to the shareholders of the acquiring firm, consequently reducing the information asymmetries between managers and shareholders of the acquirer (Bergh et al., 2019).

Motivated by the previously described gap in existing literature, the research question addressed is:

What is the impact of combined acquirer information signaling on cross-border M&A performance?

Two information signals are investigated in detail, these being the choice of payment, and the presence of private equity (PE) funding. Each serve as a potent signal, communicating nuanced information about the rationale and potential outcomes of an M&A transaction to shareholders of the acquiring firm.

Payment type is generally considered to be a signal of managerial confidence in a takeover (Campbell et al., 2016). Cash payments are regarded as a sign of positive management expectations of the deal, leading to positive announcement returns and thus, positive M&A performance (Campbell et al., 2016; Faccio & Masulis, 2005). This is because cash payments typically simplify the valuation process, as such reducing information asymmetries between the target and acquirer (Boeh, 2011; Travlos, 1987). Ultimately, cash payment communicates good news about the deal to the acquirer's shareholders (Schijven & Hitt, 2012).

Private equity funding of the acquiring firm additionally serves as an effective information signal. Due to the prior experience and social networks of PE investors, information asymmetries between target and acquiring firms are reduced when PE investors are present in the acquiring firm (Humphery-Jenner et al., 2017). The underlying reason is that PE investors are often actively involved in the management and governance practices of their holdings (Aggarwal et al., 2011; Bodie et al., 2024; Erel et al., 2022). Consequently, when M&A deals are PE-funded, this signals an increased likelihood of future value creation to the acquirers' shareholders, leading to positive M&A performance (Humphery-Jenner et al., 2017).

A first notable contribution to the existing literature lies in replicating similar studies using different data samples and research contexts, thereby enhancing the generalizability of prior research findings (Druckman et al., 2011; Findley et al., 2021). Although the impact of payment method and PE funding have been previously investigated, replication holds particular significance in research. By persistently exploring specific relationships within different research compositions, the external validity of past findings can be enriched (Druckman et al., 2011; Findley et al., 2021). Consequently, this research paper aims to provide a measure of the robustness and reliability of conclusions drawn from research conducted across different methodologies (Druckman et al., 2011; Findley et al., 2021).

Secondly, this research sheds light on the growing need to integrate existing research findings into a broader, comprehensive framework, as suggested by various scholars (Cartwright

& Schoenberg, 2006; Haleblan et al., 2009; King et al., 2004; King et al., 2021). Specifically, it addresses the need for research to explore interactions among previously established performance drivers to assess their relative importance (Haleblan et al., 2009; King et al., 2004; King et al., 2021). As such, it builds on previous findings to advance M&A research, providing a more holistic understanding of M&A performance dynamics and facilitating the development of comprehensive theoretical frameworks (Haleblan et al., 2009; King et al., 2004; King et al., 2021).

From a managerial perspective, this research delineates the general concept of information signaling in the world of M&A. It prompts consideration of which signals, and what combinations thereof, most effectively enhance M&A performance. Additionally, it raises further questions about the optimal implementation and communication of such signals to shareholders in real-world scenarios. Consequently, this research aims to inspire decision-makers to develop a deeper understanding of the strategic tools available to ensure value-enhancing M&A strategies.

Using a sample of 465 cross-border M&A deals gathered from the LSEG Workspace over the period January 2010 – December 2022, the impact of payment type, PE funding, and combinations of the two signals on cross-border M&A performance is investigated. To do so, an event study is employed, in which M&A performance is measured by cumulative abnormal returns (CARs) around the announcement date.

Based on various ordinary least squares (OLS) regressions, cash payment is shown to weakly significantly increase M&A returns relative to stock payment, with an impact of 3.064% after controlling for confounding variables. Private equity funding, however, is not found to statistically significantly impact CAR. Moreover, the interaction of the two signals combined generally has a negative impact on M&A performance, although this effect is statistically insignificant. In addition, various robustness tests further confirm these findings, suggesting that combined signaling does not have a significant impact on M&A performance.

The next Section describes findings from an extensive review of existing literature. Section three covers data and methodology, describing the sampling process and research methods employed. Section four subsequently reports and discusses the research results. Finally, Section five details an overview of conclusions drawn from this research paper.

2. Literature Review

In this Section, I systematically examine relevant literature related to the research topic, covering the most crucial theoretical concepts, as well as empirical findings from prior studies in the field. First, an assessment of the current state of M&A research is presented, highlighting the need for further investigation of M&A performance. Subsequently, this chapter delves deeper into the specific aspects of information signaling, including theories of adverse selection and information asymmetries. Ultimately, two common acquirer signals, payment type and private equity funding, are examined in detail. Finally, theories from the field of behavioral finance are used to describe the expected effect of combinations of signals. Throughout the entire literature review, previous empirical findings are critically analyzed and synthesized into one cohesive framework.

2.1 M&A Performance

The empirical literature on the performance of M&A transactions presents a highly inconsistent landscape, with a multitude of contradicting results. However, despite the wealth of such conflicting findings, a general notion in this field of research is that many acquisitions fail to generate significant positive value. For instance, in their extensive review of extant M&A research, Haleblan et al. (2009) conclude that the majority of studies find no evidence of value creation for acquiring firms and, in some instances, even indicate value destruction. Intriguingly, however, most studies report positive returns for targets, along with positive combined returns to targets and acquirers (Haleblan et al., 2009). Similarly, King et al. (2004) note the inconclusive state of M&A performance research in their meta-analysis, ultimately concluding a lack of positive M&A outcomes to acquirers on average. A more recent meta-analysis by King et al. (2021), incorporating additional performance predictors, finds that M&A performance does not deviate significantly from zero. Adding to the skepticism, Calipha et al. (2010) state a failure rate of approximately fifty percent. Although individual research papers may offer conflicting conclusions, presenting findings of positive, negative, and zero returns, overarching research reviews consistently emphasize the absence of value creation from takeover activity.

Professional literature further supports this belief of poor M&A outcomes. In fact, various professionals have estimated the failure rate to range from seventy to ninety percent of total acquisition activity (Christensen et al., 2011; Hitt et al., 2009; Kenny, 2020; Martin, 2016; Reuer,

2005; Selden & Colvin, 2003). These experts argue that, on average, takeovers yield little to no value for the acquiring firm's shareholders (Reuer, 2005; Selden & Colvin, 2003). In most cases, however, they are reported to be value destructive (Reuer, 2005; Selden & Colvin, 2003). Frequently cited factors contributing to M&A underperformance include a strategic mismatch between the target and acquirer, overpayment manifested through excessively large bid premiums, and inadequacies in post-acquisition integration of the two firms (Christensen et al., 2011; Hitt et al., 2009; Kenny, 2020). In addition, Sirower and Weirens (2022) underscore the significance of the announcement day as key indicator of M&A performance. Information disclosed in the takeover announcement serves as the foundation for investor sentiment, which subsequently determines the market's initial reaction to the announcement (Hamlin, 2022; Sirower & Weirens, 2022).

Altogether, the collective findings call for further investigation of M&A performance to uncover additional performance drivers or reveal the underlying conditions that render previously identified drivers most impactful (Cartwright & Schoenberg, 2006; Haleblian et al., 2009; King et al., 2004; King et al., 2021). By doing so, scholars may be able to identify how a minority of acquisitions enable value creation, thereby paving the way for future transactions to replicate success (Haleblian et al., 2009). The substantial and yet unexplained variance in M&A performance across transactions strongly indicates moderation by variables yet to be discovered, driving the need for continued exploration of potential antecedents of M&A performance (Cartwright & Schoenberg, 2006; King et al., 2004; King et al., 2021). As articulated by Cartwright and Schoenberg (2006), the existing body of research on M&A performance is "incomplete in some way," failing to capture the true determinants of takeover performance. What is more, King et al. (2004) recommend new research to make use of antecedents cited in prior papers to construct novel theoretical frameworks. Furthermore, the authors propose for future research to investigate interactions between antecedents as to shed light on relationships among antecedents (King et al., 2004; King et al., 2021). The conclusion drawn from these research reviews is clear: current literature lacks a holistic, integrated understanding of performance drivers in acquisitions (Cartwright & Schoenberg, 2006; Haleblian et al., 2009).

2.2 Information Asymmetry, Adverse Selection, and Information Signaling

Within the field of M&A research, numerous classifications and distinctions exist, including the differentiation between domestic and cross-border or international M&A. Cross-border M&A, which accounts for roughly thirty percent of total takeover activity, is increasingly popular due to the growing globalization of markets (Chari, 2020; Erel et al., 2022). Consequently, this form of M&A presents interesting research opportunities yet to be explored. While many aspects of the deal remain the same, substantial differences exist between domestic and cross-border M&A activities. For instance, in contrast to domestic takeovers, cross-border M&A may be driven by different factors, with the most obvious one being the desire to expand to international markets (Erel et al., 2012; Erel et al., 2022). Additionally, a favorable regulatory or political landscape may serve as motivation to acquire internationally (Erel et al., 2012; Erel et al., 2022). However, differences between countries introduce an additional layer of complexity to cross-border deals, potentially impeding the performance of these transactions (Erel et al., 2012; Erel et al., 2022). Such country-level differences generally lead to information asymmetries between the target and acquiring firms (Boeh, 2011; Moeller & Schlingemann, 2005). Examples of country-level differences include differing cultures, legislative systems, and management practices (Boeh, 2011). In addition to differences in information between buyer and seller, information asymmetries between the acquiring firm's management and its shareholders are particularly relevant in M&A transactions (Bergh et al., 2019; Song et al., 2021). These asymmetries arise due to managers of acquiring firms obtaining private information about the target firm during selection and due diligence processes (Song et al., 2021). Information asymmetries, along with associated adverse selection problems, are commonly cited to be a primary deterrent of performance, particularly true in the context of cross-border M&A (Erel et al., 2012; Erel et al., 2022; Gomes et al., 2013; Tanna et al., 2021). As a result, a large portion of cross-border M&A research aims to discover how to reduce these issues (Boeh, 2011; Erel et al., 2022).

Various finance and economics scholars have extensively documented theories of information asymmetries. Key contributors include Akerlof (1970), Allen and Faulhaber (1988), Barsanneti and Camargo (2022), Gale (2001), Grinblatt and Hwang (1989), Leland and Pyle (1977), Michaely and Shaw (1994), Stiglitz (2002), and Welch (1989). In these papers, adverse selection problems are often referred to as 'lemon problems', emphasizing the challenge buyers face in distinguishing

between sellers of good and bad quality (Akerlof, 1970; Leland & Pyle, 1977; Michaeli & Shaw, 1994). One commonly cited approach to reducing information asymmetries is signaling, or information signaling (Gale, 2001; Michaeli & Shaw, 1994; Stiglitz, 2002). For example, Akerlof (1970) suggests that guarantees, brand-name goods, and licensing may efficiently signal quality. In most of these works, signaling refers to an information-revealing action taken by the seller (Stiglitz, 2002). In contrast, information revealed through the actions of the buyer are called self-selection or screening models (Stiglitz, 2002). However, in the context of mergers and acquisitions, signaling efforts of both acquiring and target firms have been investigated. In this paper, information signaling is therefore defined as any firm-level or deal-level characteristic of either the target or acquiring firm that may signal the quality of the investment to shareholders of acquiring firms. Hence, effective information signals convey information about the quality of the deal to the shareholders of the acquiring firm, consequently reducing the information asymmetries between managers and shareholders of the acquirer (Bergh et al., 2019).

Within the realm of cross-border M&A research, various signaling efforts have been previously investigated in detail. One such signal is the payment type of the takeover. Typically, acquiring firms have the choice between cash, stock, or a combination of both as modes of payment (Faccio & Masulis, 2005). Empirical findings on payment type consistently indicate negative M&A performance for stock-financed deals (Faccio & Masulis, 2005). Conversely, cash payments typically correspond with positive M&A announcement returns (Campbell et al., 2016). These findings remain robust across various sample periods and geographical locations (Agrawal et al., 1992; Giannopoulos et al., 2017; Schijven & Hitt, 2012; Travlos, 1987; Walker, 2000).

Building upon these empirical findings, the theoretical underpinnings of payment type as a signaling mechanism in M&A shed further light on the rationale behind these observed findings. Payment type functions as an information signal, serving as a proxy for the deal expectations of the acquiring firm's management (Campbell et al., 2016). Specifically, cash payment is considered a sign of managerial confidence in the transaction (Campbell et al., 2016). On the other hand, stock payments signal an inability to capitalize on future value creation (Campbell et al., 2016; Schijven & Hitt, 2012). The predominant rationale behind this is that stock payments involve a transfer of risk to the target's shareholders, whereas cash payments do not involve such a transfer of risk (Campbell et al., 2016; Martin, 1996; Schijven & Hitt, 2012). Because of this, cash financing

simplifies the payment process, thereby reducing information asymmetries between the target and acquirer regarding the acquirer's value (Boeh, 2011). Cash payment is considered to be less complex, since its valuation is straightforward compared to that of stock payment (Boeh, 2011). Moreover, cash payment is typically preferred when the acquiring firm is undervalued, whereas stock payment tends to be favored when the acquiring firm is overvalued (Agrawal et al., 1992; Travlos, 1987). As a result, announcements of cash deals are perceived by the acquirer's shareholders as good news about the acquirer's value, whereas announcements of stock deals are interpreted as bad news (Travlos, 1987). Thus, the payment method effectively communicates information about management's confidence in the deal to the acquiring firm's shareholders, subsequently shaping investors' perception of the deal (Schijven & Hitt, 2012).

In short, based on previous empirical findings and the underlying theoretical reasoning, the first hypothesis is constructed:

H1: cash payment positively impacts cross-border M&A performance relative to stock payment.

Additionally, the presence of institutional investors represents another previously researched form of information signaling in mergers and acquisitions. For instance, Ferreira et al. (2010) study the influence of foreign institutional ownership in acquiring and/or target firm on individual and combined returns. Their findings reveal that this presence is associated with more successful M&A transactions, particularly in countries with weak legal institutions and underdeveloped markets (Ferreira et al., 2010). The authors furthermore attribute this impact to the facilitating role that institutional investors play in corporate takeovers, as such reducing information asymmetries (Ferreira et al., 2010). Similarly, Humphery-Jenner et al. (2017) find that private equity (PE) funding of acquiring firms is positively associated with returns in cross-border M&A deals if target firms are in informationally challenging environments. According to their findings, PE funding enhances M&A performance due to the prior experience and social networks that PE investors may possess in target countries, thus mitigating information asymmetries between target and acquiring firms (Humphery-Jenner et al., 2017). Consequently, PE-funded deals may signal an increased likelihood of future value creation, which is subsequently reflected in the announcement returns (Humphery-Jenner et al., 2017).

More specifically, private equity investors are widely recognized for their active involvement in the management of their investment companies (Bodie et al., 2024; Erel et al., 2022). Additionally, institutional investors exert influence over corporate governance practices of their portfolio holdings (Aggarwal et al., 2011). Notably, PE funds often secure seats on the board of directors and leverage this position to effect operational and strategic enhancements of their investments (Bodie et al., 2024; Erel et al., 2022). Professional literature further underscores this notion, asserting that private equity investors are required to enact operational improvements through direct engagement to be able to generate returns (Agnew et al., 2024). Lastly, an increasing trend towards shareholder activism suggests an even stronger active participation of PE funding going forward (Anabtawi & Stout, 2008; Haleblan et al., 2009). The significance of prior acquisition experience and social networks has been documented extensively in M&A research, highlighting what enables PE investors to positively engage in their investment holdings. Although findings are somewhat mixed, there is a prevailing consensus that both factors are relevant for improved M&A outcomes (Barkema & Schijven, 2008; Gomes et al., 2013; Haleblan et al., 2009; Meglio et al., 2023; Vermeulen & Barkema, 2001). As such, the past experience and social ties of private equity shareholders in the acquiring firm may be leveraged, alongside their proactive engagement, to increase M&A performance. Consequently, the presence of PE funding signals deal quality to the acquirer's shareholders, as such reducing the information asymmetries between management and shareholders of acquiring firms.

Based on this discussion of past empirical and theoretical literature, the following hypothesis is formulated with regards to this second form of information signaling:

H2: PE funding of acquiring firms positively impacts cross-border M&A performance relative to no PE funding.

2.3 Combinations of Information Signals

As mentioned earlier in the literature review, there is an evident need for further examination of M&A performance drivers, particularly by integrating extant knowledge. Several review articles emphasize the relevance of interactions among previously identified antecedents (King et al., 2004; King et al., 2021). Haleblan et al. (2009) further illustrate that empirical research adopts a reductionist approach. As a result, existing literature has predominantly investigated M&A

performance drivers in isolation, wherein the direct effects of individual antecedents are primarily examined (Haleblian et al., 2009). One promising way of addressing this gap and integrating extant knowledge involves evaluating the relative significance of previously identified performance antecedents (Haleblian et al., 2009). Campbell et al. (2016) specifically note the lack of knowledge on ‘complex’ signals, explaining that the aggregation of signals and its impact on investors’ announcement response has received little to no attention by academics so far. As such, an interesting research proposal emerges: the exploration of combined value drivers to understand how their collective impact differs from the sum of their individual contributions. In the context of information signaling, the field of M&A research may benefit from investigation of combinations of acquirer signals as opposed to isolated, individual signals.

In expectation, combinations of acquirer information signals may impact M&A performance differently compared to the aggregate of their individual effects. This can be argued for by applying theories of investor behavior as part of the behavioral finance literature. Effective information signals within corporate takeover announcements are expected to indicate value creation to the acquirer’s shareholders, or at the very least, a lack of value destruction (Humphery-Jenner et al., 2017). In practice, however, the initial reaction of shareholders to takeover announcements is often influenced by investor sentiment (Hamlin, 2022; Sirower & Weirens, 2022). Faced with multiple signals simultaneously, investors may disproportionately react to the announcement due to irrationality. Bodie et al. (2024) distinguish between information processing errors and behavioral biases. Similarly, Jordan et al. (2021) categorize cognitive errors and behavioral biases. The extensive body of literature on behavioral finance has elucidated numerous irrationalities capable of shaping investor sentiment. Rather than listing all possibly applicable biases, this section will highlight select examples.

First of all, investors are known to exhibit limited attention toward newly distributed information (Bodie et al., 2024). As a consequence, they often tend to so-called heuristics, or rules of thumb, when making decisions (Bodie et al., 2024; Jordan et al., 2021). Combining two or more acquirer signals may hinder investors’ ability to thoroughly evaluate and incorporate the information provided by all signals into their market response. Hence, limited attention may lead to disproportionate announcement returns when integrating multiple acquirer information signals.

Another potentially pertinent cognitive bias is confirmation bias, which prompts investors to selectively focus on information that aligns with their prior beliefs while ignoring conflicting information (Bodie et al., 2024). In the context of M&A transactions, shareholders of acquiring firms may disproportionately react to information that confirms their optimistic outlook regarding the deal's potential for future value creation, while overlooking signals suggesting value destruction associated with the acquisition. This tendency to seek confirmation of preconceived notions can result in a skewed interpretation of multiple signals.

Drawing upon the concept of investor irrationality, combinations of acquirer information signals are evidently expected to impact M&A performance differently compared to the sum of their individual impacts. Nevertheless, the direction of the effect is unclear, as different forms of investor biases may have differing consequences. As such, combined signals may exhibit either weaker or stronger impacts on M&A performance.

Following this discussion of investor irrationality and its influencing role on combined signals, the last hypothesis is developed:

H3: the combined impact of payment type and PE funding of acquiring firms on cross-border M&A performance differs from the sum of the individual impacts.

3. Methodology and Data

3.1 Data Sample

To collect data, LSEG Workspace is used. This database contains information on worldwide M&A transactions, including deal-level and firm-level characteristics. Deals are filtered to include only completed deals in which a hundred percent of shares were acquired (e.g. Campbell et al., 2016; Moeller et al., 2004). Furthermore, to ensure availability of data on the acquirer's investor base and stock prices, acquirers must be public. The sample includes only cross-border deals. To further narrow the search, deal value is required to be equal to or higher than one million USD (e.g. Moeller et al., 2004). The sample includes transactions completed between January 2010 and December 2022. Finally, certain deal types are excluded, such as spinoffs, recapitalizations, repurchases, privatizations, and self-tenders (Boeh, 2011). The initial sample consisted of 11,770 observations. However, 3,976 observations are lost due to their payment types differing from the chosen measurement. A further 3,958 and 3,346 observations are dropped due to missing acquirer size and bid premium estimates. Lastly, eleven observations missed a Datastream code.

To supplement this dataset, stock price data for all acquiring firms and a market portfolio is gathered using LSEG Datastream, for a period of 250 trading days before until one trading day after the announcement date. Fourteen more observations are dropped due to missing stock price data in the estimation window. As such, the final sample consists of 465 unique deals.

3.2 Methodology

The hypotheses will be tested using ordinary least squares (OLS) estimators as part of a cross-sectional multivariate regression analysis, as is frequently employed in acquisition research (Brooks, 2019; Datta et al., 2001; Defrancq et al., 2021; Erel et al., 2012). As part of this analysis, p-values are used to determine the significance of the regression estimates. The results tables will indicate significance levels as follows: *** for $p < .01$; ** for $p < .05$; and * for $p < .10$. A p-value smaller than the respective threshold results in rejection of the null hypothesis at the strong, moderate, or weak significance level. The regression models corresponding with the previously established hypotheses are as follows:

1) $CAR_i = \beta_0 + \beta_1 * Payment_i + \varepsilon_i$ where CAR_i = cumulative abnormal return over day -1 to day +1 relative to the announcement date; $Payment_i$ = one for cash payments, zero for stock payments; β_1 = the impact of cash payment on CARs relative to stock payment; β_0 = the expected value of CAR with stock payment; and ε_i = error term

2) $CAR_i = \beta_0 + \beta_1 * PE_Funding_i + \varepsilon_i$ where $PE_Funding_i$ = one for PE-funded acquirers and zero otherwise; β_1 = the impact of PE funding on CARs relative to no PE funding; and β_0 = the expected value of CAR without PE funding.

3) $CAR_i = \beta_0 + \beta_1 * Payment_i + \beta_2 * PE_Funding_i + \beta_3 * (Payment_i * PE_Funding_i) + \varepsilon_i$ where $Payment_i * PE_Funding_i$ = the interaction term, which is equal to one for deals with cash payment and PE funding and zero otherwise; β_3 = the interaction effect, capturing how the combined impact of cash payment and PE funding differs from the sum of their individual impacts on CAR; β_2 = the impact of PE funding on CARs relative to no PE funding and stock payment; β_1 = the impact of cash payment on CARs relative to stock payment and no PE funding; and β_0 = the expected value of CAR with stock payment and no PE funding.

3.3 Variables¹

Dependent Variable

The dependent variable is M&A performance. However, there is no universally accepted definition for this, as its measures are highly dependent on the research question (Das & Kapil, 2012). These measures can be classified into several themes. For instance, M&A performance encompasses metrics indicative of value creation for the target, acquirer, or for the two firms combined (Das & Kapil, 2012). Additionally, measures of M&A performance can diverge based on phases within the M&A process, such as pre- versus post-takeover (Calipha et al., 2010). Finally, measures distinguish based on short-term versus long-term definitions (Agrawal et al., 1992).

In this research, M&A performance is measured using cumulative abnormal returns (CARs). This market-based measure is most suitable as the research question concerns the impact of combined

¹ A full list of all key variable definitions, sources, and relevant formulas is detailed in Appendix A

acquirer information signaling on cross-border M&A performance. Information signals are typically disclosed during the announcement phase; hence, stock market reactions present the most pertinent metric for assessing the M&A performance corresponding to those signals. This aligns with established literature, as market-based measures are the most widely employed measures of M&A performance in finance research, particularly within the framework of an event study (Agrawal et al., 1992; Campbell et al., 2016; Das & Kapil, 2012; Halebian et al., 2009; King et al., 2004; Kusewitt, 1985). Event studies evaluate stock market responses to events related to a corporate takeover (Das & Kapil, 2012). CARs represent the unexpected change in stock prices around an event, and as such capture investor reactions to newly released information (Campbell et al., 2016). CAR measures assume that share price changes reflect the true value of a takeover (Das & Kapil, 2012; Kusewitt, 1985). Therefore, acquirer CARs serve as the optimal measure for discerning the implications of combined acquirer signaling on M&A performance in this thesis.

To specify this measure further, the event window will be a three-day window, starting one trading day prior to the announcement date and ending one trading day past the announcement date. Short-term horizons using daily stock returns are most precise, as they increase the statistical power of the study and avoid the occurrence of confounding events (Agrawal et al., 1992; Bhagat & Romano, 2002; Campbell et al., 2016; Das & Kapil, 2012; McWilliams & Siegel, 1997). For takeover announcements, one-to-five-day event windows are typically used (King et al., 2021). In order to separate the event's return impact from that of other, unrelated price fluctuations, estimates of normal or expected returns are needed (Agrawal et al., 1992; Brooks, 2019). In accordance with prior literature, the estimation window for the expected returns is a 200-trading day window (Brooks, 2019; Humphery-Jenner et al., 2012; McWilliams & Siegel, 1997). This will be from 249 trading days until 50 trading days before the announcement date. To calculate expected returns, the most widely used method, a basic market model, is used (Brooks, 2019). During the estimation window, daily returns of acquiring firms will be regressed on a constant and the return on a global market portfolio, the MSCI All Country Index, to obtain the firm's beta (Brooks, 2019). Subsequently, the expected return for acquiring firms during the event window is calculated by multiplying the firm's beta with the observed market return that day (Brooks, 2019). To prevent issues with extreme observations of alpha estimates, the observed alphas from the regressions are not incorporated in the calculation of normal return (Brooks, 2019). Instead, the expected value of alpha for all firms is zero (Brooks, 2019).

The use of CAR metrics as the predominant measure of M&A performance in existing literature is well-established, with several benefits. The primary motivation for utilizing this measure is its inherently forward-looking nature; CARs represent the expected value of future cash flows generated by the deal on the announcement date (Das & Kapil, 2012). In contrast, numerous other measurements, such as accounting metrics, are backward-looking, without adequately isolating the effect of the acquisition itself (Das & Kapil, 2012). Accounting measures may also be susceptible to changes in accounting standards and are often subjective due to the influence of management on accounting practices (Kusewitt, 1985; McWilliams & Siegel, 1997).

Independent Variables

The two independent variables included in this study are 1) payment type, and 2) private equity funding. In past research, various forms of measurements are used for payment type. For instance, Campbell et al. (2016) operationalize payment type as a cash ratio, referring to the percentage of cash used to finance the deal. More commonly, however, scholars have used dummy variables to indicate payment options. Schijven and Hitt (2012), for example, assign values of one, two and three for payments of cash, combinations, and stock. Other academics using similar scales include Faccio and Masulis (2005), Hayward and Hambrick (1997), and Travlos (1987). Important to note is that in these definitions, cash payment strictly refers to payment fully in cash while stock payment refers to payment fully in stocks. In this study, payment type is operationalized using a dummy variable, which is coded as zero for stock payments and one for cash payments. This approach is consistent with prior research and corresponds with the discussion of payment methods in the previous Section, in which only the impacts of cash versus stock payments are discussed.

The second independent variable, PE funding, is similarly measured using a dummy variable. PE funding refers to the presence of private equity investors in the acquiring firm's ownership. To measure this, a dummy variable is employed, where a value of one indicates the presence of PE funding and zero indicates its absence. This is in line with methodologies used in previous literature, for instance as done by Humphery-Jenner et al. (2017) in their investigation of PE backing of acquiring firms.

Control Variables

In order to isolate the effect of the independent variables on the dependent variable and reduce endogeneity concerns, control variables are included based on variables incorporated in similar studies performed in the past (Schijven & Hitt, 2012).

First, the size of the acquiring firm is controlled for, since the assets, capabilities and resources of acquirers may impact M&A performance (Datta et al., 2001; Song et al., 2021; Tanna et al., 2021; Vermeulen & Barkema, 2001). According to King et al. (2021), the size of acquiring firms is a significant predictor of M&A success, for various measures of M&A success. As such, acquirer size should be controlled for. Datta et al. (2001) and Moeller et al. (2004), among others, find a so-called size effect: announcement returns for acquirers are typically higher for smaller acquirers. Acquirer size is measured by the market capitalization, calculated as the total shares outstanding times the acquirer's stock price four weeks before the announcement date (Datta et al., 2001; Goergen & Renneboog, 2004).

Second, the acquisition premium is controlled for (e.g. Goergen & Renneboog, 2004; King et al., 2021; Song et al., 2021; Travlos, 1987). Bid premium is often considered to be an important determinant of acquisition performance (Haleblian et al., 2009; Kusewitt, 1985). For instance, Campbell et al. (2016) describe how the acquisition premium may serve as a signaling mechanism, where high premiums represent managerial confidence in value creation. Bid premium is defined as the deal value in excess of the target firm's market value four weeks before announcement, as a percentage of the target firm's market value four weeks before the announcement date (Datta et al., 2001; Laamanen, 2007; Travlos, 1987).

A third control variable is deal relatedness, referring to the similarity of the target and acquiring firms in terms of business operations (King et al., 2021). Generally, related takeovers are more likely to create value than unrelated takeovers (Scanlon et al., 1989; Walker, 2000). Deals are considered related when the primary two-digit SIC codes of target and acquiring firm are the same (Goergen & Renneboog, 2004; Scanlon et al., 1989; Walker, 2000).

Finally, to control for any macro-economic changes, year dummies are used based on the year during which the announcement took place (Boeh, 2011; Schijven & Hitt, 2012). Industry-fixed effects are captured using industry dummies based on the acquirer's SIC code.

3.4 Descriptive Statistics

The yearly deal volume, detailed in Figure 1, exhibits a range from a minimum of 20 deals in 2020 to a peak of 47 deals in 2012 and 2017. A substantial portion of all deals in the sample (61.51%) involves target firms based in the United States or Canada, as depicted in Figure 2. Similarly, nearly half of the acquiring firms (47.53%) is from the United States, Canada, or the United Kingdom, as illustrated in Figure 3. A complete list of deal frequencies by target and acquirer countries is found in Appendix B.

Figure 1. Deal Volume per Year

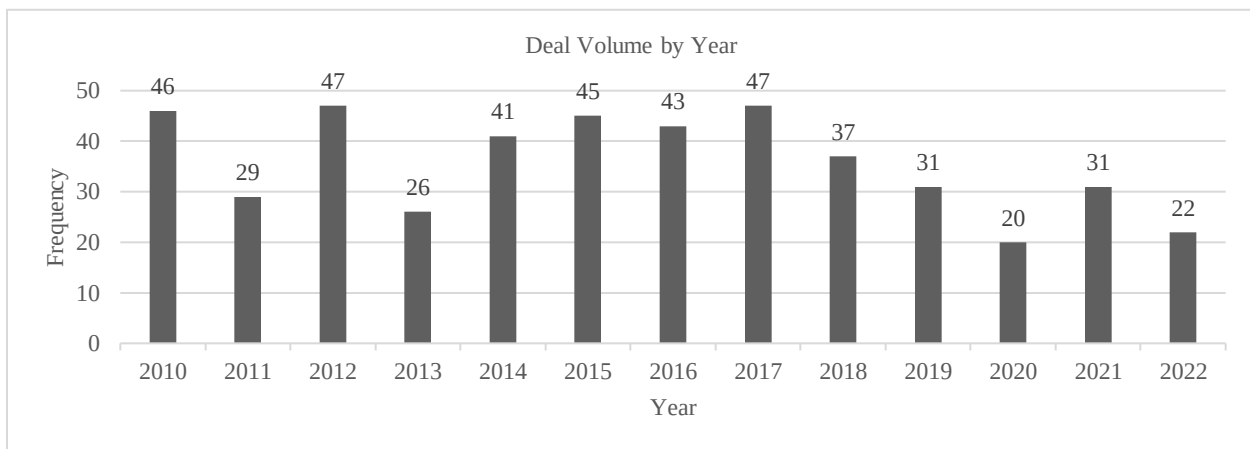


Figure 2. Top Five Target Countries

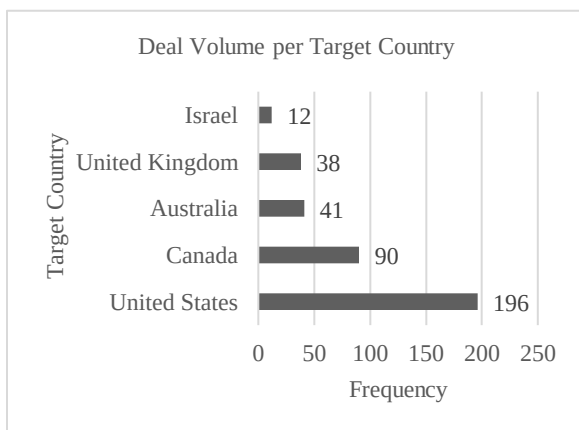


Figure 3. Top Five Acquirer Countries

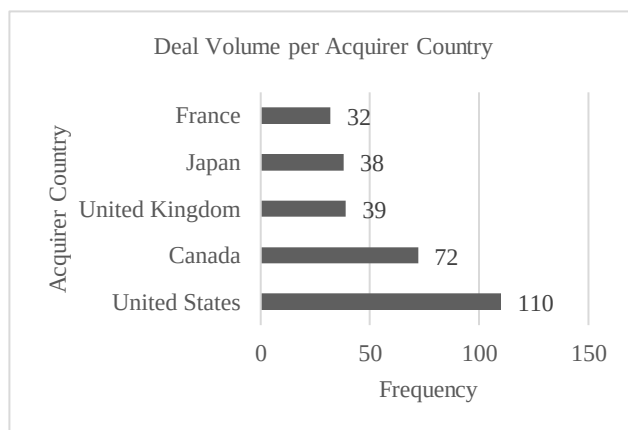


Table 1 reports summary statistics for the continuous variables included in this study. Average cumulative abnormal return is .279%, with a large standard deviation of 6.396%. In comparison, Humphery-Jenner et al. (2017) find mean CAR of 2.165% in their investigation of PE backing and M&A returns.

The mean size of acquiring firms is 27,935 million USD. This finding is in line with prior literature; Goergen and Renneboog (2004), for instance, report market capitalization of 21,568 million USD. Variability in acquirer size is notably high, with a standard deviation of 98,979 million USD.

The mean bid premium is 50.08%. In comparison, Laamanen (2007) finds an average bid premium of 53.5%. The minimum bid premium of -79.64% suggests that the sample includes deals sold at a discount.

Table 2 reports summary statistics for the dummy variables included in this study. A large portion of the transactions, 61.72%, is unrelated. This finding is in line with that of other scholars. For instance, Humphery-Jenner et al. (2017) find a mean value of diversifying or unrelated transactions of 0.667 or 66.7%.

A notably small portion of the sample is PE-funded, a mere 4.30%. Comparably, Humphery-Jenner et al. (2017) report mean PE backing of 0.018 or 1.8%.

Most deals in the sample, 343, are cash-financed, whereas only 122 deals are stock-financed. This finding is in line with results from prior research. For instance, Faccio and Masulis (2005) report 80% cash deals.

Table 1. Descriptive Statistics of Continuous Variables

Variable	Mean	Standard Deviation	Min	Max	Winsor	Skewness	Kurtosis
CAR (%)	.279	6.396	-17.701	22.300	1% level	.29	4.83
Acquirer Size (million USD)	27,935	98,979	2.68	1,935,970		15.70	298.73
Bid Premium (%)	50.08	50.54	-79.64	299.09	1% level	1.64	9.46

Table 2. Descriptive Statistics of Dummy Variables

Variable	Frequency (0)	Frequency (1)	Percentage (0)	Percentage (1)
Deal Relatedness	287	178	61.72%	38.28%
PE Funding	445	20	95.70%	4.30%
Payment	122	343	26.24%	73.76%

4. Results and Discussions

4.1 Empirical Analysis

In order to test the three hypotheses stated in Section 2, OLS regressions are estimated. For each hypothesis, two separate models are reported; the first model includes only the explanatory variable(s) as stated in the hypothesis along with the dependent variable, CAR, while the second model additionally incorporates the control variables described in Section 3. To adjust for heteroskedasticity and to furthermore account for clustering of events, standard errors are robust and clustered by the acquiring firm using the acquirer's Datastream code. This is applied to all regression models. Additionally, to alleviate problems with regards to skewness and kurtosis, the natural logarithm of acquirer size is used in all regressions (Datta et al., 2001). Furthermore, to mitigate the impact of outliers, the bid premium and CAR variables are Winsorized at a 1% level.

Table 3 provides results of OLS regressions to investigate the average impact of payment type on M&A performance. In both Model 1 and 2, the coefficient of Payment is positive, with values of 1.246% and 3.064%. This finding suggests outperformance of cash payment relative to stock payment, as expected according to H1 and in line with findings from prior research (e.g. Faccio & Masulis, 2005). This effect, however, is statistically insignificant in Model 1 ($p = .165$), while only weakly statistically significant in Model 2 ($p = .051$). The intercept terms are inconsistent across the two regressions, with expected values of CAR in case of stock payment of $-.640\%$ and $.813\%$ in Model 1 and 2, consecutively. Nevertheless, neither of these findings are statistically significant, with p-values of $.452$ and $.747$. As expected (e.g. Datta et al., 2001; Moeller et al., 2004), the coefficient for Acquirer_Size_Log is negative ($-.356\%$), although insignificant ($p = .174$). In terms of economic significance, this impact is minimal, as a 1% increase in acquirer size will lead to a $-.004\%$ change in CAR. Bid premium negatively impacts CAR by $-.002\%$, as opposed to the positive effect predicted by Campbell et al. (2016). Consequently, the impact of a one-standard-deviation increase in bid premium on CAR is $50.54 * -.002 = -.101\%$. Once again, however, this coefficient is statistically insignificant ($p = .773$). Deal relatedness positively impacts CAR by $.678\%$ relative to unrelated deals, yet this too is statistically insignificant ($p = .478$). Year-fixed effects do not seem to collectively explain the variance in CAR, as their joint significance exceeds the 5% cut-off point ($p = .569$). Industry-fixed effects, on the other hand, appear to collectively explain M&A performance, with a p-value of $.000$. Overall, the explanatory power of Model 1 is

extremely low, with an R-squared of .74%. However, the explanatory power of Model 2 is improved, with an R-squared of 44.87% and an adjusted R-squared of 8.96%. In general, these values indicate that factors other than the variables included in these models may predominantly influence M&A performance. Based on the findings from these regression models, there is weak statistically significant evidence in support of H1; there is limited indication of a significant relationship between payment method and CAR.

Table 3. Announcement CAR: Payment as Signal

Dependent variable	CAR (in %, [-1, 1])	
	Model 1	Model 2
Payment	1.246 (.165)	3.064* (.051)
Acquirer_Size_Log		-.356 (.174)
Bid_Premium_Wins		-.002 (.773)
DealRelatedness		.678 (.478)
Intercept	-.640 (.452)	.813 (.747)
Year dummies		Yes
F-statistic (year dummies)		F (12, 398) = .88 (.569)
Industry dummies		Yes
F-statistic (industry dummies)		F (92, 398) = 446.35*** (.000)
R-squared	.0074	.4487
Adjusted R-squared		.0896
Sample size	465	465

OLS regressions in Model 1 and Model 2 are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Model 2, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** p < .01, ** p < .05, * p < .10.

Table 4 provides OLS regression results to explore the effect of private equity funding on CAR. In Model 3, the coefficient for PE_Funding is -.526%, indicating that PE-funded acquirers on average obtain lower announcement returns than their counterparts, which contradicts the expected direction stated in H2 and the findings from Humphery-Jenner et al. (2017). Interestingly, Model 4 suggests the opposite; here, the coefficient is equal to 1.818%, suggesting higher returns to PE-funded acquirers. Furthermore, neither coefficient is found to be statistically significant, with both p-values largely exceeding the significance levels ($p = .652$ and $p = .338$). The intercept term in Model 3 is equal to .302%, indicating the expected value of CAR for non-PE-funded acquirers. The power of these models to explain the variance in takeover CARs is again low, with R-squared of .03% in Model 3 and adjusted R-squared of 6.70% in Model 4. Based on these two regressions, there is insufficient evidence to suggest a statistically significant relationship between PE funding of acquiring firms as an information signal and announcement returns. As such, I find no support for H2.

Table 4. Announcement CAR: PE Funding as Signal

Dependent variable	CAR (in %, [-1, 1])	
	Model 3	Model 4
PE_Funding	-.526 (.652)	1.818 (.338)
Acquirer_Size_Log		-.062 (.808)
Bid_Premium_Wins		.000 (.971)
DealRelatedness		.156 (.871)
Intercept	.302 (.343)	.865 (.742)
Year dummies		Yes
F-statistic (year dummies)		F (12, 398) = .83 (.616)

Table 4. Continued

Dependent variable	CAR (in %, [-1, 1])	
	Model 3	Model 4
Industry dummies		Yes
F-statistic (industry dummies)		F (92, 398) = 1199.99*** (.000)
R-squared	.0003	.4350
Adjusted R-squared		.0670
Sample size	465	465

OLS regressions in Model 3 and Model 4 are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Model 4, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** $p < .01$, ** $p < .05$, * $p < .10$.

Table 5 shows the results of two regression models used to determine the effect of combined information signaling on announcement returns. Similar to the findings in Models 1 and 2, payment type is found to have a positive coefficient (1.343% and 3.409%). In Model 6, the impact of payment type is moderately significant ($p = .037$), suggesting that cash payment evidently has a positive impact on M&A performance for non-PE-funded deals. In contrast to the output from Models 3 and 4, PE_Funding has a positive coefficient in both models (1.371% and 6.673%), suggesting higher returns to acquirers with PE funding compared to those without, given stock payment. Nevertheless, these coefficients are also statistically insignificant ($p = .777$ and $p = .170$). Hence, based on these models, incorporating one of the two signals on average leads to higher announcement CAR relative to the case of not incorporating either signal. The interaction term capturing the combined impact of cash payment and PE funding is negative in both regressions, with coefficients of -2.481% and -6.253%. This suggests that the combination of cash payment and PE funding as information signals on average leads to lower announcement returns than in the base case, with stock payment and without PE funding. However, this finding is statistically insignificant in both models (.614 and .220). As such, these regressions provide insufficient statistical evidence in support of H3. The R-squared of Model 6 is the highest so far, with 45.47% of the variance in CAR captured by this regression model.

Table 5. Announcement CAR: Payment and PE Funding as Signal

Dependent variable	CAR (in %, [-1, 1])	
	Model 5	Model 6
Payment	1.343 (.143)	3.409** (.037)
PE_Funding	1.371 (.777)	6.673 (.170)
Payment * PE_Funding	-2.481 (.614)	-6.253 (.220)
Acquirer_Size_Log		-.345 (.187)
Bid_Premium_Wins		-.000 (.906)
DealRelatedness		.551 (.565)
Intercept	-.685 (.429)	.490 (.849)
Year dummies		Yes
F-statistic (year dummies)		F (12, 398) = .89 (.559)
Industry dummies		Yes
		F (93, 398) = 2745.37*** (.000)
R-squared	.0087	.4547
Adjusted R-squared	.0023	.0931
Sample size	465	465

OLS regressions in Model 5 and Model 6 are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Model 6, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** $p < .01$, ** $p < .05$, * $p < .10$.

4.2 Robustness Checks²

Event Window

To check the reliability of the findings, a first robustness check employs a different event window. Instead of a three-day window, a five-day window is used, with CARs calculated from two trading days before until two trading days after the announcement date (Schijven & Hitt, 2012).

Table 6 reports the findings from all six regression models using a five-day window. The direction of the effect of cash payment on M&A performance remains robust, with positive coefficients for Payment of 1.723% and 3.176% in Model 7 and 8, respectively. This effect is statistically significant at the 10% level in both Models ($p = .087$ and $p = .072$). As such, I find some evidence in support of H1. Similar to before, none of the coefficients in Model 9 and 10 are statistically significant, providing no evidence of PE funding functioning as an effective signal. Moreover, the interaction term remains negative in this set of regression models, with coefficients equal to -5.210% and -5.600%, although these are statistically insignificant ($p = .328$ and $p = .352$).

Payment

As highlighted in Section 3, the measurement for payment type includes deals paid fully in stock and fully in cash. This measurement greatly restricts the sample size, with over 3,000 deals lost as a result. As such, another robustness test employs a different measurement. In these regressions, payment is captured with a dummy variable of three levels: 0 = cash and stock combination; 1 = stock only; and 2 = cash only. As a consequence, the sample size increases from 465 to 541 deals.

Table 7 reports the corresponding results. Notably, cash payment is positive and statistically significant at the 1% significance level in Model 13 and 14, providing strong evidence of outperformance of cash payment relative to payments with combinations of cash and stock. Stock payment, however, is not shown to impact CAR statistically significantly. Moreover, the intercept in Model 13 of -2.335% is statistically significant at the moderate level ($p = .014$), indicating negative M&A performance for deals with combined cash and stock payment. Like the previously reported findings, PE funding does not appear to have a significant impact on CAR. The same is true for all interaction terms. Interestingly, the impact of acquirer size is statistically significant in Model 14 and 18, for the first time. Altogether, the findings suggest that the payment method may better be measured using three instead of two categories.

² The definitions of newly introduced variables are detailed in Appendix C

Table 6. Robustness Check 1: Announcement CAR using three-day windows

Dependent variable	CAR (in %, [-2, 2])					
	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Payment	1.723* (.087)	3.176* (.072)			1.914* (.062)	3.481* (.058)
PE_Funding			-.291 (.819)	1.016 (.647)	3.752 (.476)	5.370 (.365)
Payment * PE_Funding					-5.210 (.328)	-5.600 (.352)
Acquirer_Size_Log		-.291 (.343)		.010 (.972)		-.283 (.355)
Bid_Premium_Wins		-.002 (.848)		.004 (.638)		.003 (.757)
DealRelatedness		.986 (.362)		.486 (.650)		.901 (.405)
Intercept	-.833 (.382)	-1.312 (.667)	.450 (.205)	-1.271 (.682)	-.956 (.324)	-1.611 (.604)
Year dummies		Yes		Yes		Yes
F-statistic (year dummies)		F (12, 398) = 1.04 (.414)		F (12, 398) = 1.02 (.430)		F (12, 398) = 1.04 (.412)
Industry dummies		Yes		Yes		Yes
F-statistic (industry dummies)		F (92, 398) = 663.79*** (.000)		F (92, 398) = 183.39*** (.000)		F (93, 398) = 464.70*** (.000)
R-squared	.0109	.4214	.0001	.4090	.0144	.4245
Adjusted R-squared		.0446		.0242	.0080	.0429
Sample size	465	465	465	465	465	465

OLS regressions in all Models are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Models 8, 10, and 12, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** p < .01, ** p < .05, * p < .10.

Table 7. Robustness Check 2: Announcement CAR using three levels for Payment

Dependent variable	CAR (in %, [-1, 1])					
	Model 13	Model 14	Model 15	Model 16	Model 17	Model 18
Stock	1.646 (.198)	.335 (.824)			1.672 (.205)	.151 (.925)
Cash	2.941*** (.003)	3.598*** (.009)			3.066*** (.003)	3.660** (.015)
PE_Funding			-.708 (.606)	.723 (.657)	.554 (.869)	.170 (.947)
Stock * PE_Funding					.868 (.878)	5.658 (.275)
Cash * PE_Funding					-1.664 (.629)	.541 (.847)
Acquirer_Size_Log		-.496** (.036)		-.234 (.325)		-.488** (.042)
Bid_Premium_Wins		-.008 (.315)		-.005 (.565)		-.007 (.378)
DealRelatedness		.740 (.387)		.212 (.808)		.627 (.465)
Intercept	-2.335** (.014)	1.871 (.467)	-.060 (.848)	2.457 (.312)	-2.407** (.014)	1.808 (.504)
Year dummies		Yes		Yes		Yes
F-statistic (year dummies)		F (12, 456) = .93 (.521)		F (12, 456) = .92 (.530)		F (12, 456) = .93 (.517)
Industry dummies		Yes		Yes		Yes
F-statistic (industry dummies)		F (102, 456) = 360.77*** (.000)		F (101, 456) = 35.89*** (.000)		F (103, 456) = 5708.09*** (.000)

OLS regressions in all Models are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Models 14, 16, and 18, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** p < .01, ** p < .05, * p < .10.

Table 7. Continued

Dependent variable	CAR (in %, [-1, 1])					
	Model 13	Model 14	Model 15	Model 16	Model 17	Model 18
R-squared	.0237	.4267	.0006	.4023	.0249	.4304
Adjusted R-squared	.0201	.1052		.0698	.0158	.1032
Sample size	541	541	541	541	541	541

OLS regressions in all Models are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Models 14, 16, and 18, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** $p < .01$, ** $p < .05$, * $p < .10$.

Buy-and-hold Abnormal Returns

By using CARs as a measure of M&A performance, this research paper inherently investigates short-term performance of corporate takeovers. However, to test whether such short-term performance endures over time, a second measure of M&A performance is tested. Buy-and-hold abnormal returns (BHAR) is measured over a period starting 30 trading days prior to announcement and ending 200 trading days after the announcement date (Bhagat & Romano, 2002; Goergen & Renneboog, 2004; Schijven & Hitt, 2012). The equation of BHAR is given as follows: $BHAR_i = \prod_t^T (1 + R_{i,t}) - \prod_t^T (1 + E(R)_{i,t})$. Measures of long-run performance may better incorporate delayed investor reactions (Schijven & Hitt, 2012).

The results of the regressions using BHAR as dependent variable are shown in Table 8. Cash payment is again shown to increase M&A performance relative to stock payment, with coefficients of 9.588% and 13.859% in Model 19 and 20. Compared to the original results in Table 3, this effect appears to be more pronounced over the long run, with moderate significance in Model 19 ($p = .022$). Lastly, PE funding and control variables are not found to have a statistically significant effect on BHAR in any of the regression models. Interestingly, in sharp contrast with the results in Table 5, the interaction terms are positive in Model 23 and Model 24, with coefficients of 12.635% and 20.386%.

Table 8. Robustness Check 3: Announcement BHAR

Dependent variable	BHAR (in %, [-30, 200])					
	Model 19	Model 20	Model 21	Model 22	Model 23	Model 24
Payment	9.588** (.022)	13.859* (.061)			9.025** (.035)	12.842* (.088)
PE_Funding			8.787 (.366)	17.985 (.258)	-1.909 (.914)	2.481 (.909)
Payment * PE_Funding					12.635 (.543)	20.386 (.457)
Acquirer_Size_Log		.259 (.860)		1.621 (.198)		.299 (.833)
Bid_Premium_Wins		-.010 (.796)		.005 (.906)		-.008 (.833)
DealRelatedness		7.240 (.201)		4.416 (.432)		6.521 (.258)
Intercept	-6.183 (.111)	9.539 (.538)	.512 (.750)	9.924 (.520)	-6.120 (.123)	10.965 (.477)
Year dummies		Yes		Yes		Yes
F-statistic (year dummies)		F (12, 398) = .82 (.629)		F (12, 398) = .70 (.755)		F (12, 398) = .77 (.677)
Industry dummies		Yes		Yes		Yes
F-statistic (industry dummies)		F (92, 398) = 22,264.07*** (.000)		F (92, 398) = 1,139.02*** (.000)		F (93, 398) = 3,339.01*** (.000)
R-squared	.0154	.3957	.0028	.3905	.0187	.4034
Adjusted R-squared		.0022		-.0064	.0123	.0078
Sample size	465	465	465	465	465	465

OLS regressions in all Models are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Models 20, 22, and 24, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** p < .01, ** p < .05, * p < .10.

Historical Averages

Lastly, a different model is used when determining the expected return needed for calculations of CAR. Instead of the market model, historical average returns are used as estimates of a firm's normal return. Such a simple approach may outperform more complicated models, since more complex models are typically accompanied by large estimation errors (Brooks, 2019). The historical average is computed using the same estimation period as in previous regressions, from 249 trading days before until 50 trading days before the announcement date.

Table 9 reports the corresponding regression results. These regressions demonstrate robustness of the previous findings, with the directions of the main effects remaining the same. For instance, Payment has positive estimators of 1.454% and 2.902% in Model 25 and 26, respectively. In line with the results in Table 3, this effect is significant at the weak level in Model 26. Additionally, the impact of cash payment on M&A performance for non-PE-funded acquirers is 3.283%, as displayed in Model 30. Nevertheless, this finding is statistically significant only at the weak significance level as opposed to the moderate significance found in Table 5. Similar to previous regressions, PE funding as well as the combination of the two signals is not found to statistically significantly impact M&A performance in this robustness check. Consequently, employing a different benchmark model in CAR calculations does not provide any further evidence in support of the three hypotheses.

Table 9. Robustness Check 4: Announcement CAR using historical averages as estimates of normal return

Dependent variable	CAR (in %, [-1, 1])					
	Model 25	Model 26	Model 27	Model 28	Model 29	Model 30
Payment	1.454 (.120)	2.902* (.074)			1.568 (.100)	3.283* (.052)
PE_Funding			-.717 (.564)	1.183 (.546)	1.489 (.777)	6.600 (.218)
Payment * PE_Funding					-2.885 (.588)	-6.984 (.208)
Acquirer_Size_Log		-.237 (.359)		.037 (.878)		-.228 (.377)
Bid_Premium_Wins		-.006 (.477)		-.003 (.656)		-.005 (.578)
DealRelatedness		.637 (.519)		.168 (.865)		.535 (.588)
Intercept	-.941 (.289)	-.043 (.986)	.163 (.621)	-.002 (.999)	-.990 (.273)	-.417 (.869)
Year dummies		Yes		Yes		Yes
F-statistic (year dummies)		F (12, 398) = .82 (.631)		F (12, 398) = .81 (.640)		F (12, 398) = .84 (.610)
Industry dummies		Yes		Yes		Yes
F-statistic (industry dummies)		F (92, 398) = 211.91*** (.000)		F (92, 398) = 38,029.05*** (.000)		F (93, 398) = 149.16*** (.000)
R-squared	.0093	.4459	.0005	.4338	.0111	.4515
Adjusted R-squared		.0851		.0651	.0047	.0878
Sample size	465	465	465	465	465	465

OLS regressions in all Models are estimated using the same sample. Standard errors are robust and clustered by acquiring firm. In Models 26, 28, and 30, year and industry dummies are included but not reported individually. Instead, joint significance of year and industry dummies is tested using F-statistics. Table reports coefficients estimates with corresponding p-values in brackets. Note: *** p < .01, ** p < .05, * p < .10.

5. Conclusion

This thesis is set out to explore the drivers of value creation in cross-border M&A deals. A consensus in the realm of M&A suggests that many takeovers fail to yield positive value. This necessitates a continued exploration of the factors influencing M&A performance, in order to discover how M&A deals can efficiently create positive value. More specifically, the central research question addressed is: "*What is the impact of combined acquirer information signaling on cross-border M&A performance?*" The comprehensive literature review highlights the importance of information asymmetry and signaling theory in M&A contexts, forming the theoretical base of the study. In light of the fact that information asymmetries are generally detrimental to M&A performance, this thesis explores how such information asymmetries can be reduced in acquisitions. By focusing on the interaction of various signals, such as payment methods and private equity involvement, this study aims to provide an understanding of how these signals collectively influence M&A performance.

The research makes significant contributions to the existing literature on M&A by addressing two crucial aspects. Firstly, through the replication of similar studies using alternative data samples and research contexts, this research enhances the generalizability of prior findings (Druckman et al., 2011; Findley et al., 2021). By persistently exploring specific relationships across varied research compositions, this research underscores the importance of robustness and reliability in drawing conclusions from M&A research conducted through different methodologies (Druckman et al., 2011; Findley et al., 2021). In doing so, this research calls into question the external validity of past findings.

Additionally, this study responds to the growing need to integrate existing research findings into a broader, comprehensive framework, as advocated by leading scholars (Cartwright & Schoenberg, 2006; Haleblan et al., 2009; King et al., 2004; King et al., 2021). By delving into the interactions among established performance drivers, the research advances the understanding of M&A performance dynamics and facilitates the development of more holistic theoretical frameworks (Haleblan et al., 2009; King et al., 2004; King et al., 2021). Consequently, this research paper addresses gaps in previous studies that typically examine performance drivers in isolation. As such, this thesis responds to scholarly calls for an understanding of how multiple information signals interact and how their relative importance predicts M&A success.

Empirically, the study employs an event study methodology to measure M&A performance through cumulative abnormal returns (CARs) surrounding the announcement date. This is supplemented by cross-sectional regression analyses to examine the determinants of M&A returns, focusing on variables such as payment type, PE funding, and acquirer size. To do so, a dataset of 465 cross-border M&A deals is used.

In line with findings from prior literature, cash payment generally leads to higher CARs compared to stock payment, indicating a positive market perception of cash-financed deals (e.g. Agrawal et al., 1992; Faccio & Masulis, 2005). This highlights the signaling effect of cash payment. However, these findings are merely weakly statistically significant. I find evidence of a strong statistically significant relationship only when measuring payment type using three instead of two categories. As such, there is some, although limited, evidence in support of a robust relationship between payment type and M&A performance.

In sharp contrast with the evidence from Humphery-Jenner et al. (2017), private equity funding is not found to significantly positively impact M&A performance. In fact, I find inconsistent evidence for the relationship between PE funding and M&A performance, with various positive and negative coefficients across regression models. On top of that, the effect of PE funding on CARs is statistically insignificant across all regression models and robustness checks. Consequently, this research paper provides no evidence to support the functioning of PE funding as an effective information signal in M&A contexts.

Lastly, an investigation of the combined impact of information signals, as captured by the interaction term, does not offer sufficient proof in support of H3. As hypothesized in Section 2, the combined impact of payment type and PE funding of acquiring firms on M&A performance is expected to differ from the sum of their individual impacts. Firstly, the OLS regressions in Section 4 do not show a consistent direction of the effect, as the interaction term is both positive and negative in various regressions. For the most part, however, the combined impact of the two signals on M&A performance is shown to be negative. In addition, the interaction term is statistically insignificant across all performed regressions. Hence, this thesis does not bring forth any evidence of the relative importance of signals in the context of M&A.

Although the combined signals are not found to statistically significantly impact M&A performance, this research paper does shed light on the general concept of information signaling in the world of M&A for managers in practice. It prompts consideration of which signals, and what combinations thereof, can most effectively enhance M&A performance. Additionally, it raises further questions about the optimal implementation and communication of such signals to shareholders in real-world scenarios. Consequently, this research seeks to inspire decision-makers to develop a deeper understanding of the strategic tools available, enabling value-creating M&A strategies. As such, this study highlights the importance of a comprehensive signaling strategy in M&A transactions for practitioners.

A first shortcoming of this research is its relatively small sample size. Despite an initial dataset containing over 10,000 observations, a significant portion of this is excluded due to missing values for key variables. In comparison to similar studies, the sample size of 465 deals appears considerably small. For instance, Humphery-Jenner et al. (2017) analyze a sample of 4,452 M&A deals in their investigation of the impact of PE funding, while Faccio and Masulis (2005) make use of a sample of 3,667 deals in their research on the impact of payment type. A small sample size poses a limitation for several reasons. For instance, small samples tend to yield more biased and less accurate coefficient estimators, consequently decreasing the explanatory power of a regression model (Brooks, 2019). Additionally, small sample sizes typically increase the likelihood of type I and type II errors (Brooks, 2019). Altogether, small samples tend to decrease the internal and external validity of a research.

Another critical limitation is associated with the event study methodology, particularly concerning the calculation of cumulative abnormal returns. Event studies rely heavily on various assumptions, and a violation of any of these may bias the regression outcomes (Brooks, 2019; McWilliams & Siegel, 1997). For instance, one inherent assumption is that individual events are independent of one another (Brooks, 2019; McWilliams & Siegel, 1997). In practice, however, events such as M&A activity often occur in waves, resulting in clustering over time and interdependence (Chari, 2020). Another drawback of event studies is that some firms may have used management discretion to strategically time announcements when this is most favorable (Brooks, 2019). Furthermore, the calculation of CARs involves various subjective, arbitrary decisions regarding the length of the estimation period, the event window, and the choice of the

model for determining normal returns (McWilliams & Siegel, 1997). Inappropriately justifying these decisions forms an issue in the research design (McWilliams & Siegel, 1997). Additionally, extreme outliers during the estimation period can substantially distort the calculation of normal returns, complicating the interpretation of test statistics (Brooks, 2019; McWilliams & Siegel, 1997). Therefore, it is essential to exercise caution when interpreting the test outcomes.

Building on the findings and limitations of this research paper, several directions for future research are evident. To address the limitations posed by the small sample size in this thesis, future research should aim to replicate this study with a significantly larger sample. By doing so, researchers can enhance the generalizability of the findings. A larger sample size improves the internal and external validity of any study, ensuring more robust and reliable conclusions.

Next, the field of M&A research can benefit from an investigation of qualitative data from case studies or interviews to provide valuable insights into the strategic considerations involved in M&A decisions (Cartwright & Schoenberg, 2006; Haleblan et al., 2009). Such alternative approaches can complement a quantitative analysis by improving the understanding of the underlying mechanisms, such as behavioral decision-making processes, that drive M&A performance (Haleblan et al., 2009).

Another promising avenue for future research is the investigation of various other measures of M&A performance. By employing an event study with performance measured in CAR, this thesis inherently focuses on a measure of short-term performance. Other, more long-term performance metrics include various accounting measures, such as return on assets (ROA) and return on equity (ROE), or other M&A outcomes such as the bid premium and organizational innovation (Das & Kapil, 2012; King et al., 2004; King et al., 2021; Kusewitt, 1985).

Most importantly, however, future research endeavors should continue the exploration of combinations of information signals. For instance, scholars can investigate the impact of interactions between more than two signals. Alternatively, upcoming research should consider other information signals, such as the presence of reputable underwriters (Wu et al., 2013). This will allow for a better understanding of how various signals collectively influence M&A performance.

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Appendix

Appendix A. Key Variable List

Table 10. Key Variables, Descriptions, and Sources

Variable	Description	Source
CAR	Cumulative abnormal return (in %) of the acquirer from one trading day before until one trading day after the announcement, estimated using betas from the market model over an estimation period from 249 trading days until 50 trading days before the announcement. Winsorized at the 1st and 99th percentile values. $CAR_i = \sum_{t=-1}^1 AR_{i,t}$ $AR_{i,t} = R_{i,t} - E(R)_{i,t}$ $E(R)_{i,t} = \beta_i R_{m,t}$	Self-constructed using price data from LSEG Workspace
P (Price)	End-of-day price	LSEG Workspace (Datastream)
R (Return)	$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$	Self-constructed using price data from LSEG Workspace
Payment	Dummy variable equal to 1 for payments fully in cash and equal to 0 for payments fully in stock	LSEG Workspace, Consideration Structure
PE_Funding	Dummy variable equal to 1 for acquirers with private equity funding and 0 otherwise	LSEG Workspace, Financial Sponsor Role
Acquirer Size	Market capitalization of the acquiring firm, measured as total shares outstanding times stock price four weeks prior to the announcement	LSEG, Acquiror Market Value 4 Weeks Prior to Announcement
Acquirer_Size_Log	The natural logarithm of Acquirer Size	Self-constructed using Acquirer Size from LSEG Workspace

Bid Premium	Excess of the purchase price per share over the target's stock price four weeks prior to the announcement, expressed as a percentage of the target's stock price four weeks prior to the announcement	LSEG Workspace, Premium Paid – 4 Weeks Prior to Announcement
Bid_Premium_Wins	A Winsorized version of Bid Premium at the 1 st and 99 th percentile values	Self-constructed using Bid Premium from LSEG Workspace
DealRelatedness	Dummy variable equal to 1 for related deals and 0 for unrelated deals, where a related deal refers to a deal for which the target and acquirer have the same primary two-digit SIC industry code	Self-constructed using LSEG Workspace, Acquiror Primary SIC and Target Primary SIC
Payment *PE_Funding	Dummy variable equal to one for deals with cash payment and PE funding and zero otherwise	Self-constructed using Payment and PE_Funding from LSEG Workspace

Appendix B. Deal Volume by Acquirer and Target Countries

Table 11. Deal Volume by Acquirer Country

Acquirer Country	Frequency	Acquirer Country	Frequency
Argentina	1	Mexico	3
Australia	17	Morocco	1
Austria	1	Netherlands	8
Belgium	6	New Zealand	1
Bermuda	7	Nigeria	1
Brazil	1	Norway	2
British Virgin Islands	1	Peru	1
Canada	72	Poland	2
Chile	1	Singapore	1
China (Mainland)	6	South Africa	7
Colombia	1	South Korea	2
Denmark	3	Spain	4
Finland	1	Sweden	13
France	32	Switzerland	18
Germany	18	Taiwan	3
Hong Kong	7	Thailand	1
India	6	Turkey	1
Ireland	12	United Arab Emirates	1
Israel	4	United Kingdom	39
Italy	4	United States	110
Japan	38	Uruguay	1
Jersey	1	Vietnam	2
Kuwait	1		
Luxembourg	1		
Malaysia	1		

Table 12. Deal Volume by Target Country

Target Country	Frequency
Australia	41
Bahamas	1
Bahrain	1
Belgium	2
Bermuda	5
Brazil	2
Canada	90
China (Mainland)	4
Denmark	2
Finland	4
France	4
Germany	1
Guernsey	3
Hong Kong	3
India	2
Ireland	1
Israel	12
Italy	2
Japan	1
Jersey	2
Luxembourg	1
Morocco	1
Namibia	1
Netherlands	10
New Zealand	3
Norway	6
Papua New Guinea	1

Target Country	Frequency
Poland	2
Singapore	3
South Africa	3
South Korea	1
Spain	3
Sweden	5
Switzerland	2
Taiwan	1
Thailand	1
Turkey	1
United Arab Emirates	1
United Kingdom	38
United States	196
Vietnam	2

Appendix C. Additional Variable List

Robustness Check 1: Event Window

$CAR_i = \sum_{t=-2}^2 AR_{i,t}$ where $AR_{i,t} = R_{i,t} - E(R)_{i,t}$ and $E(R)_{i,t} = \beta_i R_{m,t}$. Normal return is estimated using betas from the market model over an estimation period from 249 trading days until 50 trading days before the announcement. Winsorized at the 1st and 99th percentile values.

Robustness Check 2: Payment

Stock = dummy variable equal to one for deals paid fully in stock and zero otherwise.

Cash = dummy variable equal to one for payments fully in cash and zero otherwise.

Stock * PE_Funding = dummy variable equal to one for deals with stock payment and PE funding and zero otherwise.

Cash * PE_Funding = dummy variable equal to one for deals with cash payment and PE funding and zero otherwise.

Robustness Check 3: BHAR

$BHAR_i = \prod_{t=-30}^{200} (1 + R_{i,t}) - \prod_{t=-30}^{200} (1 + E(R)_{i,t})$ where $E(R)_{i,t} = \beta_i R_{m,t}$. Normal return is estimated using betas from the market model over an estimation period from 249 trading days until 50 trading days before the announcement. Winsorized at the 1st and 99th percentile values.

Robustness Check 4: Historical Average

$CAR_i = \sum_{t=-1}^1 AR_{i,t}$ where $AR_{i,t} = R_{i,t} - E(R)_{i,t}$ and $E(R)_{i,t} = \mu_i$. Normal return is estimated using historical averages over an estimation period from 249 trading days until 50 trading days before the announcement. Winsorized at the 1st and 99th percentile values.